Predicting Multidimensional Research Impact: A Neurosymbolic Approach using LLMs and Impact-Augmented Knowledge Graphs

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Abstract. Traditional metrics such as citation counts and attention indicators often fail to capture the multidimensional and evolving nature of research impact. This paper introduces Flowmetrics, a neurosymbolic framework that conceptualises impact as a structured progression through five semantic stages: Reach, Engagement, Feedback, Influence, and Outcome. The framework is operationalised through an impact-augmented knowledge graph that combines arXiv metadata with platform-level attention signals, including tweets, citations, and policy mentions. We investigate whether large language models (LLMs) can infer and order impact trajectories between topic pairs based on this structured evidence. Seven LLMs are evaluated across three axes: stage-level classification, sequence-level coherence, and cross-model agreement. Results show that a subset of models, particularly Claude 3 Sonnet, GPT-4o, and Mixtral, produce consistent stage predictions and coherent trajectories that reflect the expected canonical structure. These models also display strong agreement in how they interpret the evidence. Overall, the findings indicate that LLMs, when guided by structured prompts, can reason about research impact in an interpretable and systematic way, supporting future metascientific applications.

Keywords: Research impact · Knowledge graphs · Large Language Models · Neurosymbolic Reasoning.

1 Introduction

Conventional citation-based metrics, such as citation counts, journal impact factors, and more recent attention-based indicators, are increasingly criticised for their narrow scope and inability to capture the multifaceted nature of research impact [6,8]. While long regarded as the gold standard for evaluating scholarly output, these metrics often neglect crucial dimensions such as societal engagement, interdisciplinary relevance, and real-world outcomes [6,30]. Consequently, they provide an incomplete and often inequitable picture of research contributions, reinforcing outdated academic incentives and systemic biases [5].
This over-reliance on citation-based indicators has led to notable distortions in the research ecosystem. For example, studies indicate that female researchers

receive fewer citations than male counterparts, even when producing equally or more innovative work, particularly in fields such as biology and medicine [36,50]. Moreover, citation-centric evaluation practices incentivise behaviours that prioritise publication volume over societal value [10]. Furthermore, as the global volume of scientific output grows, these traditional metrics' inadequacies become increasingly evident.

Simultaneously, the landscape of research dissemination has undergone significant transformation. Preprint platforms, open-access repositories, and scholarly social networks such as ResearchGate and Mendeley have expanded the visibility of research, facilitating interactions that extend well beyond citation trails [7,11]. Altmetrics have emerged to capture early indicators of attention, including social media mentions, online shares, policy citations, and public discussions, offering complementary insights into how research resonates with diverse audiences [34,6]. These developments underscore that research is now encountered through different channels and that attention manifests in multiple forms that are not reflected by conventional metrics.

What remains absent from existing metrics is any representation of how different forms of attention relate to one another. They offer snapshots — counts of citations, tweets, or policy mentions — but provide no indication of what these interactions signify or how they relate within a broader pattern of attention [49,29]. A citation, a social media mention, and a policy reference are treated as equivalent units, even though they reflect fundamentally different forms of attention [34,15]. Capturing these differences requires analysing attention at the level of broader research areas, where interactions cluster around topics and often emerge at the intersections between them [9]. Collapsing such distinct attention signals into single counts flattens how research gains visibility, engages different communities, or contributes to outcomes in professional or public settings [5,43]. Addressing these limitations requires moving beyond static tallies towards representations that characterise attention as a set of connected interactions — metrics understood not as isolated numbers, but as components of a structured pattern of attention [45,31].

In response, **Flowmetrics** models research impact as a structured progression through five semantic stages: *Reach*, *Engagement*, *Feedback*, *Influence*, and *Outcome*. The framework is operationalised through an *impact-augmented knowledge graph* that integrates arXiv metadata with platform-level attention signals such as tweets, citations, and policy mentions. The graph captures both semantic relationships between research topics and the associated evidence of their dissemination. Large Language Models (LLMs) are prompted with subgraphs and metadata to infer which impact stages are supported for a given topic pair and to reason about plausible stage sequences [44,25,1].

This study evaluates whether LLMs can infer and order research impact trajectories from structured evidence. We assess seven models across three axes: (i) stage-level classification with knowledge graph evidence, (ii) sequence-level coherence and canonical ordering, and (iii) cross-model agreement in stage pre-

dictions. Together, these evaluations offer insight into the reasoning capabilities of LLMs over structured representations of impact.
To guide this investigation, we address the following research questions:

- **RQ1**: Can LLMs reliably infer impact stages between research topics from structured evidence?
- **RQ2**: To what extent do LLMs produce coherent and canonically ordered impact trajectories?
- **RQ3**: How consistent are impact stage predictions across different LLM families?

In summary, our contributions are: (i) we construct an impact-augmented knowledge graph that models research topics and their associated forms of impact; (ii) we evaluate seven LLMs across stage-level, sequence-level, and cross-model dimensions; (iii) we analyse model performance in terms of accuracy, coherence, and consistency; and (iv) we release the codebase and data pipeline to support reproducibility and future work.

2 Literature Review

2.1 Theoretical Foundations: Diffusion of Innovations

Our development of Flowmetrics draws conceptually on Everett Rogers' *Diffusion of Innovations* theory [35], which models how new ideas and technologies spread over time through communication channels within a social system. Rogers identifies five key elements that shape this diffusion process: (i) the innovation itself, (ii) communication channels, (iii) time, (iv) the social system, and (v) adopter categories, such as innovators, early adopters, early majority, late majority, and laggards.
Building on this staged perspective, Flowmetrics conceptualises research impact as a set of five semantic stages: visibility (*Reach*), interaction (*Engagement*), response (*Feedback*), change in discourse (*Influence*), and uptake (*Outcome*). These stages reflect how research transitions from initial exposure to tangible societal effects, with attention signals serving as proxies for each stage, even though Flowmetrics does not yet model temporal diffusion explicitly.

2.2 Foundations of the Flowmetrics Impact Stages

Each Flowmetrics stage is grounded in a distinct body of theoretical and empirical work, providing a structured lens for interpreting how research impact unfolds.
Reach captures initial visibility and access to research, drawing on altmetrics and media studies that quantify exposure via views, downloads, and mentions on platforms like Twitter [34,2].
Engagement denotes active interaction with research, including commenting, sharing, and discussion. It is informed by social network theory and community-based dissemination models [6,46].

Feedback involves reflexive responses such as critique and refinement. It builds on co-production frameworks and iterative models of knowledge formation, including peer review and dialogic engagement [23,18,12].
Influence reflects a topic's contribution to scholarly discourse, typically measured through citations and bibliometric indicators [13,40,21].
Outcome refers to the broader societal effects of research, such as its impact on policy, practice, or innovation. It is informed by frameworks like the Theory of Change and Responsible Research and Innovation (RRI) [51,42].
We assume these stages collectively define a multidimensional model of research impact that conceptually spans both epistemic and societal forms of attention.

2.3 Scientific Knowledge Graphs

The construction of structured representations of scholarly knowledge has gained increasing attention through the development of Scientific Knowledge Graphs (SKGs) [37]. These graphs encode entities such as papers, authors, institutions, and research topics, and their relationships, including citations, authorship, collaboration, and topical similarity.
The *Microsoft Academic Graph (MAG)* [48] was one of the earliest large-scale SKGs, integrating scholarly metadata with hierarchical field-of-study taxonomies. Its successor, *OpenAlex* [33], provides an open, continuously updated alternative that preserves key MAG features while offering greater transparency and access.
Other initiatives emphasise semantic enrichment. The *Open Research Knowledge Graph (ORKG)* [17] captures machine-actionable research contributions through structured statements, enabling comparison and synthesis of findings. Linked Data approaches such as *ScholarlyData* [32] and the RISIS RDF dataset model scholarly objects and events using ontologies and RDF.
The *Semantic Scholar Open Research Corpus (S2ORC)* [28] provides citation graphs augmented with semantic features, supporting large-scale machine learning on scientific text. More recently, the *AIDA Knowledge Graph* [3] links research papers to a fine-grained representation of their research topics, the types of their authors' affiliations, and associated industrial sectors.
Additionally, the *dblp Knowledge Graph* [24] offers a rich, structured view of computer science literature, capturing metadata such as authors, venues, affiliations, and co-authorship networks. While such efforts provide valuable insights into the structure of scholarly communication, they are primarily centred on bibliographic entities and do not characterise the different forms of impact associated with research topics.
These limitations motivated the design of our Impact-Augmented Knowledge Graph. Unlike existing SKGs, which primarily organise documents, authors, or citation links, this graph treats research topics—and their pairwise relationships—as first-class nodes and incorporates both structural relations and platform-level attention signals. Its design is conceptually informed by diffusion-oriented perspectives that emphasise how different channels and forms of atten-

tion shape the visibility, interpretation, and uptake of research, complementing rather than relying solely on bibliographic indicators.
Although the current implementation is limited in scale, it serves as a proof of concept demonstrating the viability of representing impact as a structured, stage-based categorisation of topic-level attention.

2.4 Knowledge Graphs for Large Language Models

Neurosymbolic systems aim to bridge the gap between the structured reasoning afforded by symbolic representations, such as knowledge graphs (KGs), and the generative fluency of LLMs. In this paradigm, KGs serve both as external memory and as structured repositories of factual knowledge, enhancing the interpretability, controllability, and contextual reasoning capabilities of LLMs [4,16]. To integrate KG information into LLMs, knowledge-enhanced approaches typically follow one of three strategies. **Path-based models**, such as CE-PR [19] and MRG [52], extract relational paths between entities to support explicit logical reasoning and guided generation. **Triple-based models**, including KEPM [14], MixGEN [41], and CNTF [47], incorporate structured facts by verbalising triples into natural language prompts or by encoding them through multi-hop attention mechanisms. **Subgraph-based models**, such as KG-BART [27], GRF [20], and JointGT [22], encode local neighbourhoods using graph neural networks (GNNs), which are then fused with language model representations to enhance context-aware generation.
These strategies demonstrate the value of KGs in supporting reasoning, grounding, and transparency in language model outputs. Flowmetrics adopts a lightweight *triple-to-text prompting* strategy: relevant triples are extracted from the graph, verbalised as short natural language statements, and injected directly into the prompt as structured evidence. This approach avoids fine-tuning, ensures transparency, and enables model-agnostic, zero-shot reasoning over the structured attention signals captured in the Impact-Augmented Knowledge Graph.

3 Methodology

This section outlines the Flowmetrics pipeline, including the research task, knowledge graph construction, the evaluated LLMs, and the structured prompt.

3.1 Task Definition

We formalise research impact modelling as a two-part task including *impact stage prediction* and *structured data-to-text generation*. The first sub-task requires the model to infer which impact stages best characterise the relationship between a research topic pair (t_A, t_B), selecting one or more stages from the Flowmetrics trajectory: *Reach*, *Engagement*, *Feedback*, *Influence*, and *Outcome*. This stage-level prediction provides a structured categorisation of how the two topics are connected across different forms of research attention. The second sub-task asks

the model to generate a short, evidence-grounded explanation summarising why those stages were selected, ensuring that the predictions are interpretable and anchored in the underlying attention signals. Although the framework includes *Feedback*, this dimension is implemented in the task but not supported by evidence, as no publicly available signals capture expert commentary or peer review. We retain it for completeness, and future work will incorporate such signals through restricted-access sources or emerging open peer review datasets [26].
Stage prediction is based on structured attention signals from Altmetric[1] and CrossRef[2], which provide platform-level evidence mapped to the following stages:

- **Reach:** Twitter, Facebook, Wikipedia
- **Engagement:** Blogs, Reddit, YouTube, Mendeley
- **Feedback:** *(unsupported)* – requires expert evaluation
- **Influence:** CrossRef citations, News mentions
- **Outcome:** Policy mentions, Patent citations

To ensure consistent reasoning, we use a standardised prompt template (Appendix A) that encourages inclusive assessment of weak or cumulative attention signals. When no metrics are available, the prompt instructs models to assign *Reach* by default, indicating minimal public accessibility.

3.2 Knowledge Graph Construction

The Impact-augmented Knowledge Graph is implemented as an RDF graph using a custom vocabulary aligned with metascientific semantics. It integrates data from **arXiv**, **Altmetric**, and **CrossRef**. arXiv provides the metadata used for topic extraction and co-occurrence modelling; Altmetric contributes attention signals from social media, news outlets, blogs, and public discussion platforms, as well as mentions in policy documents and patents; and CrossRef supplies citation information from the scholarly literature. The resulting graph is built over a curated corpus of 12,350 arXiv computer science papers (2000–2024), selected for topical diversity, metadata completeness, and coverage on at least one attention platform.
The corpus spans major subfields of Computer Science, including *Computer Vision* (24.6%), *Machine Learning* (21.6%), *Computation and Language* (11.2%), and *Artificial Intelligence* (8.9%), ensuring that the graph captures variation across subject areas.
Our data model builds on SKOS[3] (`skos:`) together with a custom *Flow* ontology (`flow:`). The primary entities are **research topics**, represented as `skos:Concept`, and **topic pairs**, represented as `flow:TopicPair`. Each topic pair is linked to evidence for the five Flowmetrics stages using properties such as `flow:hasReachImpact`, which point to blank nodes encoding

[1] Altmetric — `https://www.altmetric.com/`
[2] CrossRef — `https://www.crossref.org/`
[3] SKOS Simple Knowledge Organization System – `https://www.w3.org/TR/skos-reference/`

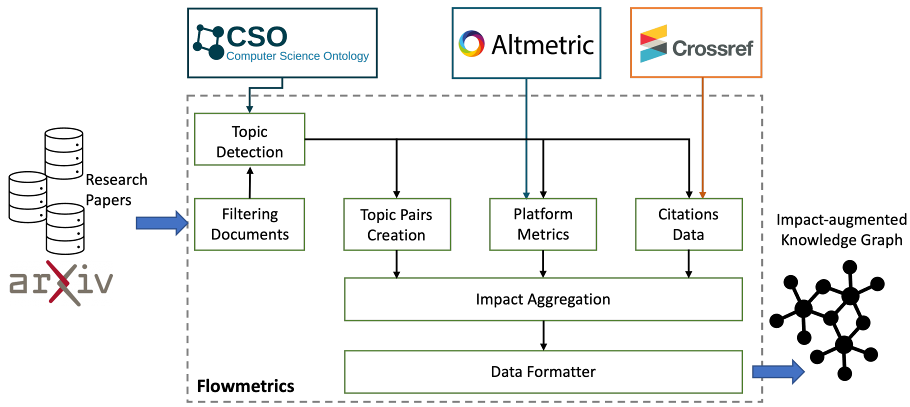

Fig. 1: Overview of the knowledge graph construction pipeline.

platform sources (e.g., Twitter, Facebook), normalised attention signals, and shared concepts between the two topics (e.g., `“language model”`). For example: `(TopicPair, flow:hasReachImpact, BlankNode)`; `(BlankNode, flow:platform, “Twitter”)`.

The final knowledge graph contains **156 topics**, **771 topic pairs**, and over **33,000 RDF triples**. The graph is anonymised, serialised in Turtle format, and released publicly to support reproducibility. Both the full `.ttl` file and an illustrative snippet for a high-impact pair (e.g., *semantics* ↔ *language model*) are available in our open repository.

Figure 1 summarises the knowledge graph construction process, showing how paper metadata, topic extraction, and platform-level attention signals are combined to produce the Impact-Augmented Knowledge Graph.

Nodes of Topics Topic nodes in the knowledge graph represent research areas extracted from arXiv metadata. We applied the *CSO Classifier* [38], an unsupervised tool that assigns concepts from the *Computer Science Ontology* (CSO) [39] based on titles, abstracts, and keywords.

The classifier combines a *syntactic module*, which detects explicitly mentioned CSO concepts, with a *semantic module* that uses part-of-speech tagging and embeddings to infer related terms. We merged both outputs to obtain candidate concepts for each paper.

To consolidate fine-grained variations, we applied a frequency-based clustering strategy, assigning each paper to its most frequently occurring concept across the corpus. This step groups similar papers under unified topic identifiers while avoiding manual curation.

The final pipeline yielded 156 topics spanning diverse areas of computer science. We verified their coherence and domain relevance by inspecting representative papers associated with the most frequent concepts and confirming alignment with the corresponding CSO labels.

Edges of Impact Edges in the knowledge graph represent impact-bearing connections between pairs of research topics. An edge is created whenever two topics co-occur in a paper and share at least one CSO concept. Each pair is modelled as a `flow:TopicPair` and linked to its evidence through RDF triples of the form:

```
(TopicPair, hasStageImpact, BlankNode)
```

where `hasStageImpact` is a stage-specific property (e.g., `flow:hasReachImpact`, `flow:hasInfluenceImpact`). Each `BlankNode` stores the contribution from one platform, identified via `flow:platform` and quantified using `flow:score`.

Aggregation Strategy. Platform-level attention signals differ substantially in scale. To make them comparable, values for each platform are first min–max normalised independently, yielding scores in $[0, 1]$. For a given topic pair, the contribution from a platform is the sum of the two normalised topic values, resulting in a platform-specific score in $[0, 2]$. One blank node is created per platform contributing to a stage. We avoid normalising by the number of platforms so that stages with richer evidence are not downweighted.

Example. Consider the pair *Semantics* and *Language Model*, which share the CSO concept `"language model"`. Their Reach-related contributions appear as:

```
(flow:pair_15_131, flow:hasReachImpact, _:reach1)
(_:reach1, flow:platform, "Twitter"), (_:reach1, flow:score, 0.1321)
(_:reach2, flow:platform, "Facebook"), (_:reach2, flow:score, 0.0972)
```

Let $v_{A,\mathrm{Tw}}$ and $v_{B,\mathrm{Tw}}$ be the raw Twitter counts for the two topics, and $[v^{\mathrm{Tw}}_{\min}, v^{\mathrm{Tw}}_{\max}]$ the range across all topics. Normalisation gives:

$$\mathrm{norm}(v_{t,\mathrm{Tw}}) = \frac{v_{t,\mathrm{Tw}} - v^{\mathrm{Tw}}_{\min}}{v^{\mathrm{Tw}}_{\max} - v^{\mathrm{Tw}}_{\min}}.$$

The platform-level score is the sum:

$$\mathrm{norm}(v_{A,\mathrm{Tw}}) + \mathrm{norm}(v_{B,\mathrm{Tw}}) = 0.1321.$$

The Facebook value (`0.0972`) is obtained in exactly the same way, using Facebook's own normalisation range.

Formal Definition. For stage d with platform set P_d, and raw value $v_{t,p}$ for topic t on platform p, the contribution of platform p to the pair (t_A, t_B) is:

$$I^{d,p}_{(t_A,t_B)} = \mathrm{norm}(v_{t_A,p}) + \mathrm{norm}(v_{t_B,p}),$$

where norm is min–max normalisation applied independently to each platform. These platform-level values are stored as structured triples in the graph.

3.3 Large Language Models

We evaluated seven LLMs selected for their strong performance in reasoning and generation, and for their suitability to our dual objective: *impact stage prediction* and *structured data-to-text generation*. LLMs are essential in this setting because no human-labelled dataset exists for impact stages between topic pairs; creating one would require extensive expert annotation across five stages. Traditional supervised models therefore cannot be trained, whereas LLMs can operate in a *zero-shot*, train-free setup and can directly interpret the structured evidence extracted from the Impact-Augmented Knowledge Graph. The selected models span both proprietary and open-source families, ensuring architectural diversity while remaining compatible with a uniform prompting setup (Appendix A). Their main characteristics are summarised in Table 1.

Table 1: Main characteristics of the evaluated LLMs.

Model	Type	Version	Ctx. Window
DeepSeek-V3	Open	deepseek-chat (v3-base)	128k
Mixtral 8x22B	Open	open-mixtral-8x22b	64k
GPT-3.5-Turbo	Proprietary	gpt-3.5-turbo-0125	16k
GPT-4o	Proprietary	gpt-4o-2024-05-01	128k
GPT-4o-mini	Proprietary	gpt-4o-mini-2024-05-01	128k
Claude 3.7 Sonnet	Proprietary	claude-3-7-sonnet-20250219	200k
Claude 3.5 Haiku	Proprietary	claude-3-5-haiku-20241022	200k

3.4 Prompt Design

We use a structured zero-shot prompt to support both *impact stage prediction* and *structured data-to-text generation*. The prompt frames stage prediction as a multi-label classification task over the five Flowmetrics stages, instructing the model to determine which stages are supported by the platform-level evidence and shared research concepts provided as input. It also asks the model to produce a short semantic summary based on the same evidence. These summaries are part of the overall task design and are released in our repository, although they are not evaluated in this paper.

Each stage is briefly defined in the prompt, and platform-level attention signals are presented in a fixed, structured format. Stages appear in canonical order to encourage coherent reasoning. The *Feedback* stage is included to preserve the full Flowmetrics space, although it is not supported by the available evidence.

All models were run with temperature = 0.1 and a maximum output length of 4096 tokens, ensuring reproducibility while allowing minimal lexical variation. Model outputs were parsed into structured JSON, from which we extracted the predicted stage set and the inferred sequence.

4 Experiments

This section outlines the experimental setup and evaluates model performance along three axes: stage-level classification, sequence-level coherence, and cross-model agreement.

4.1 Setup

We evaluate seven LLMs (Table 1) using the impact-augmented knowledge graph (Section 3.2), which contains 771 topic pairs linked to platform-level attention signals across the five Flowmetrics stages. Each model receives the same structured zero-shot prompt (Section 3.4), supporting stage prediction and data-to-text generation. All experiments were run at temperature = 0.1 using a unified LangChain pipeline to ensure consistent execution across models.

4.2 Stage-Level Classification

We begin by evaluating how effectively each LLM identifies the impact stages supported by the evidence for a given topic pair. The goal is to assess whether models can correctly interpret the structured platform-level attention signals and map them to the appropriate Flowmetrics stages. Accurate stage identification is essential because it underpins all subsequent analyses: if a model cannot reliably determine which stages are present, any higher-level reasoning about impact trajectories becomes less meaningful.

Evaluation Setup. Because no human-labelled dataset exists, we derive reference labels using a simple heuristic: a stage is present for a topic pair when its cumulative evidence score exceeds 0.003. We tested several thresholds (0.0, 0.002, 0.005, 0.01, 0.02). A value of 0.0 makes some stages ubiquitous (e.g., *Reach* and *Influence* at 100%), while higher cut-offs quickly suppress plausible weak signals (e.g., *Outcome* drops to 57.1% at 0.01 and 42.2% at 0.02). Thresholds around 0.003–0.005 yielded the most stable distributions, filtering noise without discarding meaningful evidence. We therefore adopt 0.003 as an empirically supported cut-off. Under this rule, *Reach* appears in 89.6% of pairs, *Engagement* in 61.2%, *Influence* in 77.6%, and *Outcome* in 75.6%; *Feedback* is always absent due to missing expert-evaluation signals.

Table 2: Global stage-level performance metrics across models.

Model	Precision	Recall	F1-Score	Accuracy
GPT-3.5-Turbo	0.660	0.325	0.378	0.611
GPT-4o	0.722	**0.696**	0.702	**0.860**
GPT-4o-mini	**0.746**	0.624	0.667	0.763
Claude 3 Sonnet	0.717	0.697	0.688	0.859
Claude 3.5 Haiku	0.753	0.564	0.613	0.793
Mixtral 8x22B	0.714	0.727	**0.715**	0.827
DeepSeek-Chat	0.700	0.515	0.527	0.730

Results. Table 2 reports global multi-label classification performance. Claude Sonnet, GPT-4o, and Mixtral achieve the strongest results, with Mixtral obtaining the highest F1-score (0.715), and GPT-4o leading in recall (0.696) and accuracy (0.860). Claude Sonnet performs comparably, with balanced precision and recall. GPT-4o-mini and Haiku perform slightly below this top tier. DeepSeek and GPT-3.5-Turbo show the largest performance gaps, with GPT-3.5 posting the lowest recall (0.325) and F1-score (0.378), indicating difficulty detecting weaker stage signals.

Stage-wise Trends. Figure 2 shows F1-scores by stage. All models perform well on *Reach*, reflecting its dense and consistent evidence base. Most models also perform reliably on *Engagement* and *Influence*, though performance drops for *Outcome*, where evidence is sparser and more heterogeneous. This drop is most pronounced for GPT-3.5 and DeepSeek. In contrast, GPT-4o, Claude Sonnet, and Mixtral maintain consistently strong results across all supported stages.

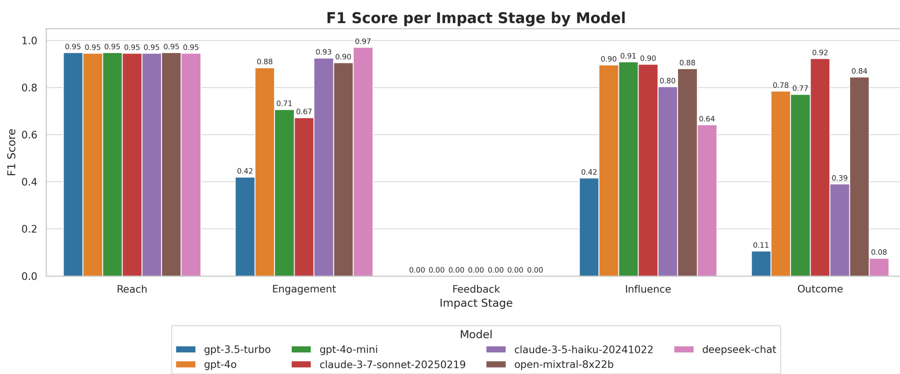

Fig. 2: Stage-level F1-score comparison across models and impact stages.

Robustness to Stage Order. We also assess how sensitive stage predictions are to the order in which stages are presented in the prompt. We evaluate 200 topic pairs across ten permutations of the stage list (details in Appendix B). All models produced valid outputs, but stability varied substantially. Claude Sonnet and Mixtral were the most robust, with mean F1 shifts of only about ±2%. GPT-4o, DeepSeek, and GPT-3.5 showed moderate sensitivity, with average changes of roughly 3–7%. GPT-4o-mini and Claude Haiku were the most affected: GPT-4o-mini exhibited deviations of up to 25%, while Haiku showed a consistent negative drift, sometimes performing 25 percentage points below the canonical condition. These results indicate that while some models rely primarily on stage semantics, others remain strongly influenced by prompt structure.

4.3 Sequence-Level Coherence

Beyond identifying which impact stages apply to a topic pair, we also assess whether models can arrange those stages into a coherent progression. Since Flowmetrics treats stages as semantically ordered categories, predictions should reflect a plausible, non-contradictory flow even when some stages are skipped.

Table 3: Model-wise sequence-level metrics (percent).

Model	Adherence (%)	Violations (%)	Coherence (%)
DeepSeek-Chat	**100.0**	31.3	69.0
Mixtral 8x22B	79.2	48.5	73.6
GPT-3.5-Turbo	97.8	30.4	60.9
GPT-4o	**100.0**	48.4	75.1
GPT-4o-mini	57.8	52.3	72.6
Claude 3 Sonnet	**100.0**	64.9	**75.6**
Claude 3.5 Haiku	**100.0**	**28.7**	69.8

Sequence-level metrics are computed only over stages that can meaningfully appear in trajectories, ensuring that coherence reflects actual evidence rather than unsupported stages.

Evaluation Setup. We evaluate sequence quality using three complementary metrics:

- **Canonical Order Adherence:** Checks whether the predicted sequence respects the expected ordering *[Reach → Engagement → Influence → Outcome]*. Skips are allowed, but reversals are not.
- **Progressive Inclusion Consistency (PIC):** Ensures that whenever a stage appears, all earlier canonical stages are also included. For example, *[Reach, Influence]* violates PIC because *Engagement* is missing.
- **Path Coherence Score:** Computes the ratio between the Longest Canonical Subsequence (LCS) and the full predicted sequence length. Higher scores indicate predictions closer to the ideal canonical path.

Results. Table 3 summarises model performance. Claude Sonnet and GPT-4o achieve perfect canonical adherence (100%) and the strongest overall coherence (75.6% and 75.1%). Haiku exhibits the lowest PIC violation rate (28.7%), suggesting stricter internal ordering. Mixtral shows robust coherence (73.6%) but struggles with PIC due to missing intermediate stages. GPT-3.5-Turbo, despite high adherence, has the weakest coherence (60.9%), reflecting unstable stage sequencing. DeepSeek is highly consistent, also achieving 100% adherence.

4.4 Cross-Model Agreement

We next examine how consistently different LLMs interpret the same structured evidence. Whereas stage-level and sequence-level evaluations assess performance relative to a proxy reference, cross-model agreement evaluates whether models converge on similar predictions when given identical inputs. High agreement indicates stable, model-independent reasoning; low agreement suggests divergent interpretations or higher variability in how models read the same evidence.

Evaluation Setup. We compute two complementary agreement metrics for every pair of models:

- **Stage Agreement (Jaccard Similarity):** measures overlap between predicted stage sets, defined as the size of the intersection divided by the union.
- **Stage Order Agreement (Kendall's Tau):** quantifies similarity in predicted stage sequences, accounting for missing stages. A value of 1.0 indicates identical ordering; 0.0 indicates no correlation.

Results. Table 4 reports pairwise agreement across all models. Claude Sonnet and GPT-4o exhibit the strongest convergence (Jaccard = 0.85, Tau = 0.86), followed closely by Mixtral (Jaccard = 0.83, Tau = 0.89). Together, these form a highly aligned cluster that produces similar stage sets and trajectories.
GPT-4o-mini aligns well with this top group (Jaccard ≈ 0.74; Tau ≈ 0.85–0.87). DeepSeek and Haiku show strong agreement with each other (Jaccard = 0.76, Tau = 0.87), forming a secondary cluster despite differing model families. GPT-3.5 displays the lowest agreement with all other models, suggesting more variable or idiosyncratic interpretations of the evidence.

Table 4: Pairwise model agreement: Jaccard similarity (upper) and Kendall's Tau (lower). Bold marks highest agreement per model.

Model	DeepSeek	Mixtral	GPT-3.5	GPT-4o	4o-mini	Sonnet	Haiku
DeepSeek	—	0.61	0.62	0.67	0.55	0.57	**0.76**
Mixtral	0.84	—	0.47	**0.83**	0.70	**0.83**	0.63
GPT-3.5	0.80	0.77	—	0.52	0.47	0.47	0.58
GPT-4o	0.84	**0.89**	0.78	—	0.76	**0.85**	0.72
4o-mini	0.74	0.79	0.77	0.85	—	0.74	0.65
Sonnet	0.71	0.81	0.74	**0.86**	**0.87**	—	0.63
Haiku	**0.87**	0.81	0.78	0.84	0.80	0.74	—

5 Discussion and Future Work

This section reflects on the experimental findings, highlights the main limitations of our approach, and outlines directions for future research.

5.1 Interpretation of Results

The experiments evaluate three aspects of model behaviour: whether LLMs can identify which impact stages are supported by structured evidence (stage-level classification), whether they can organise those stages into plausible trajectories (sequence-level coherence), and whether different models interpret the same evidence in similar ways (cross-model agreement). These analyses address the research questions posed in this paper.

At the stage level, the key question is whether models can interpret platform-level attention signals and map them to the appropriate Flowmetrics stages. Sonnet, GPT-4o, and Mixtral perform strongest overall, recovering evidence-supported stages with relatively balanced precision and recall. GPT-3.5 struggles to detect weaker or later-stage signals, and DeepSeek tends to underpredict *Outcome*. The robustness tests reinforce these patterns: models with weaker stage performance also show larger F1 shifts when stage definitions are permuted, indicating greater dependence on prompt structure rather than underlying evidence. Only a subset of models produce stable predictions under both the canonical and reordered prompts.

Sequence-level evaluation examines whether models can arrange the selected stages into coherent progressions following the Flowmetrics ordering. Sonnet, GPT-4o, Mixtral, and Haiku generally produce plausible trajectories, though with varying stability. Coherence must be interpreted separately from classification accuracy: a model may order its chosen stages sensibly but still be unstable in which stages it selects. This is visible in our results, where weaker stage-level models such as GPT-4o-mini and GPT-3.5 also produce less reliable trajectories.

Cross-model agreement evaluates whether models converge on similar interpretations of the same structured input. Sonnet, GPT-4o, and Mixtral form the most consistent group, often selecting similar stage sets and producing comparable orderings. DeepSeek and Haiku show mutual alignment but reflect a more simplified reading of the evidence. GPT-3.5 diverges most from the others, indicating higher variability in how it interprets the signals.

Overall, the findings show that only a few models behave consistently across the three evaluation axes. Claude Sonnet is the most balanced across stage classification, sequence coherence, and model agreement. GPT-4o also performs strongly but is more sensitive to prompt structure, and Mixtral remains competitive with some variability. Other models succeed on individual axes but do not provide the stability needed for reliable modelling of impact trajectories.

5.2 Limitations and Future Work

The impact-augmented knowledge graph provides a structured proxy for evaluation, but it is not a definitive ground truth. Mapping platform-level signals to Flowmetrics stages involves interpretive choices and may miss latent or delayed forms of impact. Platform biases may also distort the evidence. Future work will incorporate human-in-the-loop validation to refine these mappings and establish expert-derived ground truth for selected cases.

The prompting strategy supports structured predictions but may introduce constraints that affect model flexibility. We plan to explore alternative prompt formats and contrastive or few-shot examples to improve robustness.

The *Feedback* stage remains in the prompt, although it is unsupported due to the absence of public expert-evaluation signals. This is a data limitation. We intend to integrate proprietary expert-evaluation data from Altmetric to enable explicit modelling of feedback-oriented interactions.

Temporal dynamics are not yet represented: all experiments use static, aggregated attention signals. Incorporating timestamped Altmetric events and longitudinal citation data will allow Flowmetrics to model how impact emerges and evolves. We also plan to test whether chain-of-thought prompting improves interpretability.
A system-level ablation analysing the contribution of individual pipeline components was also conducted. Due to space limits, results are excluded from the appendix, but the full experiment is available in our public repository and supports the robustness patterns reported in the paper.
Finally, although experiments focus on Computer Science, Flowmetrics is domain-agnostic. Extending it to other fields will require adapting topic-extraction methods and validating platform relevance. Future work will also evaluate the second sub-task—structured data-to-text generation—to assess whether models can verbalise impact reasoning faithfully.

6 Conclusion

This paper introduced *Flowmetrics*, a neurosymbolic framework that moves beyond static citation or attention counts by modelling research impact as a structured progression through five semantic stages. Motivated by the limitations of conventional indicators and the fragmented nature of online attention, Flowmetrics offers a way to represent how different signals relate within a coherent trajectory of impact.
Using an impact-augmented knowledge graph and a unified zero-shot prompting setup, we evaluated seven large language models on stage classification, sequence coherence, and cross-model agreement. Claude Sonnet, GPT-4o, and Mixtral were the most reliable across evaluations, while other models showed greater variability or sensitivity to prompt structure.
Overall, the results indicate that LLMs can reasonably interpret structured impact evidence and recover key elements of the cumulative logic underlying Flowmetrics. In contrast to snapshot metrics that treat citations, tweets, and policy mentions as disconnected counts, Flowmetrics models how these signals relate within a broader pattern of attention. This study provides an early step toward representing research impact as a connected set of stages, supporting future metascientific applications and enabling more transparent and structured approaches to assessing research impact.

Resource Availability Statement: All resources, including source code, the impact-augmented knowledge graph, prompt templates, and arXiv metadata, are available at `https://anonymous.4open.science/r/predicting-multidimensional-research-impact/`.

Supplemental Material Statement: Materials to reproduce the results, including source code, prompt templates, evaluation scripts, and data, are also available at the same URL. No additional materials are required for verification.

References

1. Aggarwal, T., Salatino, A., Osborne, F., Motta, E.: Large language models for scholarly ontology generation: An extensive analysis in the engineering field. arXiv preprint arXiv:2412.08258 (2024)
2. Ahmed, Y.A., Ahmad, M.N., Ahmad, N., Zakaria, N.H.: Social media for knowledge-sharing: A systematic literature review. Telematics and informatics **37**, 72–112 (2019)
3. Angioni, S., Salatino, A., Osborne, F., Recupero, D.R., Motta, E.: Aida: A knowledge graph about research dynamics in academia and industry. Quantitative Science Studies **2**(4), 1356–1398 (2021)
4. Bhuyan, B.P., Ramdane-Cherif, A., Tomar, R., Singh, T.: Neuro-symbolic artificial intelligence: a survey. Neural Computing and Applications **36**(21), 12809–12844 (2024)
5. Bornmann, L.: What is societal impact of research and how can it be assessed? a literature survey. Journal of the American Society for information science and technology **64**(2), 217–233 (2013)
6. Bornmann, L.: Do altmetrics point to the broader impact of research? an overview of benefits and disadvantages of altmetrics. Journal of informetrics **8**(4), 895–903 (2014)
7. Bornmann, L., Haunschild, R., Mutz, R.: Growth rates of modern science: a latent piecewise growth curve approach to model publication numbers from established and new literature databases. Humanities and Social Sciences Communications **8**(1), 1–15 (2021)
8. Bornmann, L., Marx, W., Gasparyan, A.Y., Kitas, G.D.: Diversity, value and limitations of the journal impact factor and alternative metrics. Rheumatology international **32**, 1861–1867 (2012)
9. Boyack, K.W., Klavans, R., Börner, K.: Mapping the backbone of science. Scientometrics **64**(3), 351–374 (2005)
10. Fire, M., Guestrin, C.: Over-optimization of academic publishing metrics: observing goodhart's law in action. GigaScience **8**(6), giz053 (2019)
11. Fortunato, S., Bergstrom, C.T., Börner, K., Evans, J.A., Helbing, D., Milojević, S., Petersen, A.M., Radicchi, F., Sinatra, R., Uzzi, B., et al.: Science of science. Science **359**(6379), eaao0185 (2018)
12. Fyfe, A., Squazzoni, F., Torny, D., Dondio, P.: Managing the growth of peer review at the royal society journals, 1865-1965. Science, Technology, & Human Values **45**(3), 405–429 (2020)
13. Garfield, E.: Citation indexes for science: A new dimension in documentation through association of ideas. Science **122**(3159), 108–111 (1955)
14. Guan, J., Huang, F., Zhao, Z., Zhu, X., Huang, M.: A knowledge-enhanced pretraining model for commonsense story generation. Transactions of the Association for Computational Linguistics **8**, 93–108 (2020)
15. Haustein, S., Bowman, T.D., Costas, R.: Interpreting "altmetrics": Viewing acts on social media through the lens of citation and social theories. Theories of informetrics and scholarly communication pp. 372–406 (2016)
16. Hitzler, P., Sarker, M.K.: Neuro-symbolic artificial intelligence: The state of the art (2022)
17. Jaradeh, M.Y., Oelen, A., Farfar, K.E., Prinz, M., D'Souza, J., Kismihók, G., Stocker, M., Auer, S.: Open research knowledge graph: next generation infrastructure for semantic scholarly knowledge. In: Proceedings of the 10th international conference on knowledge capture. pp. 243–246 (2019)

18. Jasanoff, S.: States of knowledge: the co-production of science and the social order (2004)
19. Ji, H., Ke, P., Huang, S., Wei, F., Huang, M.: Generating commonsense explanation by extracting bridge concepts from reasoning paths. arXiv preprint arXiv:2009.11753 (2020)
20. Ji, H., Ke, P., Huang, S., Wei, F., Zhu, X., Huang, M.: Language generation with multi-hop reasoning on commonsense knowledge graph. arXiv preprint arXiv:2009.11692 (2020)
21. Jurgens, D., Kumar, S., Hoover, R., McFarland, D., Jurafsky, D.: Measuring the evolution of a scientific field through citation frames. Transactions of the Association for Computational Linguistics **6**, 391–406 (2018)
22. Ke, P., Ji, H., Ran, Y., Cui, X., Wang, L., Song, L., Zhu, X., Huang, M.: Jointgt: Graph-text joint representation learning for text generation from knowledge graphs. arXiv preprint arXiv:2106.10502 (2021)
23. Latour, B.: Science in action: How to follow scientists and engineers through society. Harvard university press (1987)
24. Ley, M.: The dblp computer science bibliography: Evolution, research issues, perspectives. In: International symposium on string processing and information retrieval. pp. 1–10. Springer (2002)
25. Liang, W., Zhang, Y., Cao, H., Wang, B., Ding, D.Y., Yang, X., Vodrahalli, K., He, S., Smith, D.S., Yin, Y., et al.: Can large language models provide useful feedback on research papers? a large-scale empirical analysis. NEJM AI **1**(8), AIoa2400196 (2024)
26. Lin, J., Song, J., Zhou, Z., Chen, Y., Shi, X.: Moprd: A multidisciplinary open peer review dataset. Neural Computing and Applications **35**(34), 24191–24206 (2023)
27. Liu, Y., Wan, Y., He, L., Peng, H., Yu, P.S.: Kg-bart: Knowledge graph-augmented bart for generative commonsense reasoning. In: Proceedings of the AAAI conference on artificial intelligence. vol. 35, pp. 6418–6425 (2021)
28. Lo, K., Wang, L.L., Neumann, M., Kinney, R., Weld, D.S.: S2orc: The semantic scholar open research corpus. arXiv preprint arXiv:1911.02782 (2019)
29. Mahrt, M., Puschmann, C.: Science blogging: An exploratory study of motives, styles, and audience reactions. Journal of science Communication **13**(3), A05 (2014)
30. Moed, H.F.: Citation analysis in research evaluation, vol. 9. Springer Science & Business Media (2006)
31. Moed, H.F.: Applied evaluative informetrics. Springer (2017)
32. Nuzzolese, A.G., Gentile, A.L., Presutti, V., Gangemi, A.: Conference linked data: the scholarlydata project. In: The Semantic Web–ISWC 2016: 15th International Semantic Web Conference, Kobe, Japan, October 17–21, 2016, Proceedings, Part II 15. pp. 150–158. Springer (2016)
33. Priem, J., Piwowar, H., Orr, R.: Openalex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. arXiv preprint arXiv:2205.01833 (2022)
34. Priem, J., Taraborelli, D., Groth, P., Neylon, C.: Altmetrics: A manifesto (2011)
35. Rogers, E.M., Singhal, A., Quinlan, M.M.: Diffusion of innovations. In: An integrated approach to communication theory and research, pp. 432–448. Routledge (2014)
36. Rossiter, M.W.: The matthew matilda effect in science. Social studies of science **23**(2), 325–341 (1993)
37. Salatino, A.A., Mannocci, A., Osborne, F.: Detection, Analysis, and Prediction of Research Topics with Scientific Knowledge Graphs, pp. 225–252. Springer International Publishing, Cham (2021). `https://doi.`

org/10.1007/978-3-030-86668-6_11, https://doi.org/10.1007/978-3-030-86668-6_11
38. Salatino, A.A., Osborne, F., Thanapalasingam, T., Motta, E.: The cso classifier: Ontology-driven detection of research topics in scholarly articles. In: Digital Libraries for Open Knowledge: 23rd International Conference on Theory and Practice of Digital Libraries, TPDL 2019, Oslo, Norway, September 9-12, 2019, Proceedings 23. pp. 296–311. Springer (2019)
39. Salatino, A.A., Thanapalasingam, T., Mannocci, A., Osborne, F., Motta, E.: The computer science ontology: a large-scale taxonomy of research areas. In: The Semantic Web–ISWC 2018: 17th International Semantic Web Conference, Monterey, CA, USA, October 8–12, 2018, Proceedings, Part II 17. pp. 187–205. Springer (2018)
40. Small, H.: Co-citation in the scientific literature: A new measure of the relationship between two documents. Journal of the American Society for information Science **24**(4), 265–269 (1973)
41. Sridhar, R., Yang, D.: Explaining toxic text via knowledge enhanced text generation. In: Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies. pp. 811–826 (2022)
42. Stahl, B.C., Obach, M., Yaghmaei, E., Ikonen, V., Chatfield, K., Brem, A.: The responsible research and innovation (rri) maturity model: Linking theory and practice. Sustainability **9**(6), 1036 (2017)
43. Sugimoto, C.R., Larivière, V.: Measuring research: What everyone needs to know. Oxford University Press (2018)
44. Thelwall, M.: Can chatgpt evaluate research quality? Journal of Data and Information Science (2024)
45. Thelwall, M., Kousha, K.: Web indicators for research evaluation. part 2: Social media metrics. Profesional de la Informacion **24**(5), 607–620 (2015)
46. Trunfio, M., Rossi, S.: Conceptualising and measuring social media engagement: A systematic literature review. Italian Journal of Marketing **2021**(3), 267–292 (2021)
47. Varshney, D., Prabhakar, A., Ekbal, A.: Commonsense and named entity aware knowledge grounded dialogue generation. arXiv preprint arXiv:2205.13928 (2022)
48. Wang, K., Shen, Z., Huang, C., Wu, C.H., Dong, Y., Kanakia, A.: Microsoft academic graph: When experts are not enough. Quantitative Science Studies **1**(1), 396–413 (2020)
49. Weller, K.: Social media and altmetrics: An overview of current alternative approaches to measuring scholarly impact. Incentives and performance: Governance of research organizations pp. 261–276 (2014)
50. Yang, A.J., Ding, Y., Liu, M.: Female-led teams produce more innovative ideas yet receive less scientific impact. Quantitative Science Studies pp. 1–21 (2024)
51. Zand, D.E., Sorensen, R.E.: Theory of change and the effective use of management science. Administrative science quarterly pp. 532–545 (1975)
52. Zhao, L., Xu, J., Lin, J., Zhang, Y., Yang, H., Sun, X.: Graph-based multi-hop reasoning for long text generation. arXiv preprint arXiv:2009.13282 (2020)

A Prompt Template

We used a single structured prompt across all models to ensure consistent task framing and accurate multi-label classification. The template includes:

- `{t_A}`, `{t_B}`: research topics being evaluated.
- `{shared_concepts}`: shared keywords extracted via the CSO Classifier.
- `{full_platform_lines}`: normalised attention signals (e.g., `Twitter: 0.132, Facebook: 0.097, Wikipedia: 0.177`).

```
You are an expert in research impact analysis. Your task is to assess the impact
↪  of a pair of research topics based on structured evidence of platform
↪  co-mentions and shared concepts.

This is a multi-label classification problem. Your goal is to classify all impact
↪  stages as either supported or not, based on the strength and relevance of the
↪  evidence for each stage.

---
Impact Stages:

- Reach: Broad dissemination of research to general audiences via mass
↪  communication platforms (e.g., Twitter, Facebook, Wikipedia).
- Engagement: Active interaction, discussion, or interpretation of research in
↪  community-driven forums (e.g., Blogs, Reddit, YouTube, Mendeley).
- Feedback: Scholarly reactions or critical appraisals, often indicating academic
↪  interest (e.g., Peer Review).
- Influence: Contribution to discourse in authoritative contexts (e.g., citations
↪  via CrossRef, media coverage via News).
- Outcome: Tangible societal or technological effects arising from research (e.g.,
↪  Policy documents, Patents).

---
Important Notes:

- Platform values are normalised per platform and impact dimension at the topic
↪  level (0.0 = no evidence, 1.0 = maximum evidence). For topic pairs, scores are
↪  summed (range: 0.0-2.0) to reflect combined impact.
- Classify all impact stages with meaningful or emerging support, considering the
↪  total cumulative evidence across platforms, even if individual platform scores
↪  are low.
- Err slightly on the side of inclusiveness: if multiple signals exist across
↪  platforms, prefer assigning the stage rather than omitting it.
- If cumulative evidence across all platforms for a stage is strictly zero, assign
↪  only the "Reach" stage.

---
Now classify this pair:

- Topic 1: {t_A}
- Topic 2: {t_B}
- Shared Concepts: {shared_concepts}

Platform Co-mention Evidence (normalised values):
{full_platform_lines}

---
You must begin your output exactly with:

Impact Stages with Sufficient Support: [list of stages]

Followed by:

Impact Summary:
Write a concise (2-4 sentence) summary explaining the evidence for each assigned
↪  impact stage. Mention the key platforms and shared concepts that support each
↪  stage. Highlight how cumulative evidence across platforms contributed to
↪  classification, even if individual signals are small.
```

B Additional Experiments

Robustness to Stage Order

To assess whether models depend on the order in which impact stages are presented in the prompt, we evaluated all seven LLMs on a stratified sample of 200 topic pairs drawn from the full KG. The sample includes a balanced mix of simpler two- to three-stage cases and more complex four-stage trajectories to ensure that the test captures behaviour across varying levels of difficulty.
Each model was run across ten permutations of the five-stage list, including the canonical ordering, the reversed list, and a diverse set of random or semantically altered permutations. Task instructions and evidence remained fixed; only the order of stage definitions was modified. For each permutation, we computed the change in F1-score (ΔF1) relative to the canonical prompt, providing a direct measure of sensitivity to stage ordering.
Models differed markedly in robustness. Claude Sonnet and Mixtral showed the smallest deviations, with mean ΔF1 values near zero (0.022 and -0.016), indicating that their predictions are largely insensitive to how the stages are listed. GPT-4o, DeepSeek, and GPT-3.5 displayed moderate shifts, with means between -0.074 and -0.028, suggesting that prompt order can influence their outputs but not dramatically. GPT-4o-mini and Claude Haiku were the most affected: GPT-4o-mini exhibited swings up to 0.25 across permutations, while Haiku showed consistently negative deviations across all permutations, showing clear sensitivity to changes in prompt order.
Table 5 reports mean, standard deviation, and extrema for each model. A full per-permutation breakdown is available in our public repository for transparency. These results show that while some models rely primarily on semantic definitions of the stages, others are more sensitive to superficial ordering, emphasising the importance of prompt stability when deploying weaker models.

Model	**Mean**	**Std**	**Min**	**Max**
Claude 3.5 Haiku	−0.201	0.072	−0.251	−0.044
Claude 3.7 Sonnet	0.022	0.053	−0.048	0.133
DeepSeek-Chat	−0.064	0.061	−0.153	0.025
GPT-3.5-Turbo	−0.028	0.090	−0.178	0.150
GPT-4o	−0.074	0.064	−0.168	0.035
GPT-4o-mini	0.062	0.116	−0.080	0.248
Mixtral 8x22B	−0.016	0.070	−0.114	0.121

Table 5: Robustness: sensitivity to stage ordering (ΔF1 relative to canonical prompt).